Question Book No: 20980
Color: GREEN

Name: _____

Roll No: _____

ENGLISH

1. **It's difficult _____ reconcile such different point of view.**

   A. with
   B. to
   C. in
   D. on

2. **I promise to _____ you in all circumstances.**

   A. Stand by
   B. Stand with
   C. Stand off
   D. Stand up to

3. **The police arrested him and his one _____.**

   A. operator
   B. colleague
   C. friend
   D. accomplice

4. **The street was littered with stones and bottles hurled _____ the policemen.**

   A. upon
   B. to
   C. on
   D. at

5. **A speed limit is the _____ legal speed that you can travel on the road.**

   A. longest
   B. highest
   C. biggest
   D. maximum

6. **He is a Bangash by _____.**

   A. Cost
   B. Cast
   C. Caste
   D. None of these

7. **Govt. must hold a/an _____ inquiry.**

   A. sensible
   B. intelligent

C. judicial
D. discreetly

8. A monkey is not capable _____ flying in the air.

A. Of
B. At
C. With
D. From

9. The president speech was so _____ that many people were persuaded to accept the need of people.

A. Expressive
B. eloquent
C. calm
D. articulate

10. If she is not interested, we will _____ the proposal.

A. vacant
B. remove
C. impulse
D. abandon

Questions 11-13 are based on the following passage.

The early 1900s saw London grappling with a persistent and hazardous problem: its infamous "pea-souper" fogs. These weren't your ordinary mists; they were thick, choking blankets of smog, a toxic blend of natural fog and industrial pollution. Coal-fired factories, power plants, and domestic chimneys belched plumes of soot and sulfur dioxide, which mingled with the moisture in the air, creating a dense, yellow-brown haze. This smog reduced visibility, disrupted transportation, and posed serious health risks, particularly for those with respiratory ailments. The Great Smog of 1952, a particularly severe episode, brought the city to a standstill and led to thousands of deaths. It served as a stark wake-up call, prompting the implementation of stricter air pollution regulations and ultimately leading to cleaner air in London.

11. What was the primary cause of London's dense fogs in the early 1900s?
A. Volcanic eruptions
B. Natural weather patterns
C. Industrial pollution
D. Excessive humidity

12. How did the Great Smog of 1952 impact London?

A. It boosted tourism due to its unique atmosphere.
B. It led to a significant improvement in air quality.

C. **It had no significant impact on the city's health or economy.**
D. **It caused widespread illness and death, prompting air pollution regulations**

13. Which of these statements is FALSE about London's fogs?

A. **They were solely caused by natural weather conditions.**
B. **They posed serious health risks to the population.**
C. **They were a common occurrence in the early 1900s.**
D. **They were a significant problem until stricter regulations were implemented.**

Choose the lettered word or phrase that is most nearly Similar in meaning to the word in capital letters.

14. DYNAMIC

A. **energetic**
B. **pure**
C. **strong**
D. **necessary**

15. DEVOID

A. **deficient**
B. **inept**
C. **invalid**
D. **endowed**

16. IMPECCABLE

A. **dirty**
B. **faultless**
C. **criminal**
D. **poverty-stricken**

17. BANISH

A. **abhorrent**
B. **frank**
C. **exile**
D. **delight**

CSS (Central Superior Services) exam in Pakistan and the UPSC (Union Public Service Commission) exam in India are both highly prestigious, competitive examinations that open the doors to civil services in their respective countries. While these exams offer the promise of power, respect, and the chance to serve the nation, they come with significant challenges and, arguably, some inherent flaws. Both exams are criticized for focusing too heavily on rote learning and theoretical knowledge rather than on practical problem-solving skills or real-world competencies. Many argue that the exams place unnecessary stress on candidates,

creating a high-stakes environment where one's future can hinge on minor errors. Moreover, the selection process often leads to elitism, where candidates from affluent, urban backgrounds, who have access to better resources, have a significant advantage over those from rural areas. Despite these criticisms, supporters of CSS and UPSC argue that these exams remain the most objective and rigorous way to select candidates for critical roles in government. They assert that the extensive preparation required fosters discipline, perseverance, and comprehensive knowledge of the country's governance, which are essential traits for a civil servant. However, this brings up a question of modernization: should these exams be restructured to focus more on assessing practical skills, emotional intelligence, and leadership qualities rather than solely academic prowess?

18. Which of the following best captures a proposed change for improving the CSS and UPSC exams?

A. Requiring a minimum of five years of work experience before eligibility
B. Emphasizing practical skills and leadership qualities over theoretical knowledge
C. Increasing the exam's difficulty level to test candidates more rigorously
D. Reducing the number of subjects to lessen the preparation burden on candidates
E. Allowing only candidates with a professional degree to appear for these exams

Choose the lettered word or phrase that is most nearly Opposite in meaning to the word in capital letters.

19. AMPLIFY

A. Embalm
B. Mummify
C. Shock
D. Curtail

20. BASE

A. Top
B. Roof
C. Climax
D. Height

21. SUBLIME

A. Low
B. Mean
C. Strange
D. Ridiculous

10. DWINDLE

A. Decay

B. **Calculable**
C. **Increase**
D. **Support**

Each question below consists of a related pair of words, followed by five lettered pairs of words. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

11. HOLY: SLOB

A. **Low: Glow**
B. **Low: One**
C. **Low: Old**
D. **Low: Bow**

12. DOLLAR: DIME

A. **Century: Decade**
B. **Retreat: Victory**
C. **Ring: Necklace**
D. **Saleswoman: Pitch**

13. HOURSE: MARE

A. **Donkey: Pony**
B. **Dog: Puppy**
C. **Fox: Vixen**
D. **Duck: Geese**

14. HOSPITAL: NURSE

A. **School: Teacher**
B. **Factory: worker**
C. **Store: Customer**
D. **Restaurant: Chef**

15. Book: Library

A. **Fish: Marine**
B. **Bird: Avian**
C. **Cat: Feline**
D. **Horse: Equine**

16. DATA: ANALYSIS

A. **Chart: Visualization**
B. **Report: Writing**
C. **Experiment: Observation**

D. Information: Decision

17. POTENTIAL: POWER

A. Energy: Movement
B. Skill: Training
C. Wisdom: Knowledge
D. Water: Thirst

18. THESIS: DISSERATATION

A. Introduction: Conclusion
B. Research: Experiment
C. Study: Survey
D. Proposal: Grant

19. CONDEMNATION: DISAPPROVAL

A. Praise: Admiration
B. Criticism: Support
C. Blame: Encouragement
D. Shock: Acceptance

Questions 32-34 are based on the following passage.

Higher Education Authorities (HEAs) play a vital role in shaping the academic, economic, and social landscapes of countries by overseeing universities, promoting research, and ensuring the quality of higher education. These bodies, such as the Higher Education Commission (HEC) in Pakistan and the University Grants Commission (UGC) in India, are responsible for policy formulation, funding allocation, and establishing standards for institutions of higher learning. Supporters argue that HEAs are indispensable for maintaining a coherent and high-quality education system; they enforce regulations, ensure curriculum relevance, and drive academic innovation. Additionally, they help maintain fair access to education by promoting scholarship programs and encouraging inclusivity. However, there are criticisms as well, particularly regarding the centralized power of HEAs. Opponents claim that these bodies often impose restrictive policies, stifling university autonomy and limiting academic freedom. Furthermore, funding limitations and bureaucratic inefficiencies in HEAs may result in a lack of timely support for research and development, ultimately hindering the nation's progress in the global academic arena. This raises a crucial question: how can Higher Education Authorities best balance regulatory responsibilities with the need to foster an environment conducive to academic freedom and innovation?

20. Which factor is essential for Higher Education Authorities to promote inclusivity in higher education?

A. Offering high salaries for faculty members
B. Encouraging private sector involvement in curriculum development
C. Providing scholarships and funding for underprivileged students
D. Reducing the number of universities
E. Enforcing strict age limits for college admission

21. Which of the following is a primary role of Higher Education Authorities?

A. **Regulating the prices of textbooks for universities**
B. **Overseeing policies and standards for higher education institutions**
C. **Limiting the number of courses offered by universities**
D. **Conducting entrance exams for primary school students**
E. **Issuing graduation certificates to all students**

22. **Which approach could help Higher Education Authorities better support innovation and academic freedom?**

A. **Increasing regulations for all university activities**
B. **Reducing the number of universities receiving funding**
C. **Granting more autonomy to universities while providing regulatory oversight**
D. **Removing scholarships for research programs**
E. **Focusing solely on undergraduate education quality**

East Africa is home to some of the world's most iconic wildlife and natural landscapes, making it a premier destination for game tourism. Countries like Kenya, Tanzania, Uganda, and Rwanda attract millions of tourists annually who come to experience safaris and observe wildlife in its natural habitat. However, the region's game tourism industry is facing several significant threats. Poaching, driven by illegal wildlife trade, is decimating populations of endangered species such as elephants and rhinoceroses. Habitat destruction due to expanding agriculture, infrastructure development, and deforestation is reducing the living space for animals, disrupting ecosystems. Additionally, climate change is altering rainfall patterns, causing droughts that affect both wildlife and tourism activities. Political instability and armed conflicts in some areas also deter tourists from visiting, leading to economic losses for local communities that depend on tourism revenue. Despite these challenges, efforts are being made to combat these threats, including anti-poaching laws, wildlife conservation programs, and eco-tourism initiatives aimed at promoting sustainable practices. However, for East Africa's game tourism to thrive in the long run, more coordinated efforts at national and international levels are required.

23. **What is one of the primary causes of population decline in endangered species in East Africa?**

A. **Expansion of agriculture**
B. **Climate change**
C. **Political instability**
D. **Poaching**

24. **Which of the following is NOT mentioned as a threat to game tourism in East Africa?**

A. **Industrial pollution**
B. **Political instability**
C. **Climate change**
D. **Habitat destruction**

25. What are some measures being implemented to protect wildlife and promote tourism in East Africa?

A. Urbanization and land reforms
B. Expanding agriculture and deforestation
C. Limiting tourist numbers and increasing taxes
D. Urbanization and land reforms

In Pakistan, taboos play a significant role in shaping societal norms and behavior, often exerting a powerful influence over individuals' lives. These taboos encompass a range of sensitive topics that are seldom openly discussed, including mental health, gender equality, sexual abuse, and caste-based discrimination. Mental health, for instance, is heavily stigmatized, with individuals who suffer from conditions like depression or anxiety often being labeled as weak or unstable, preventing many from seeking professional help. Similarly, discussions about sexual harassment and abuse remain largely silenced, with victims fearing social backlash or being blamed for their experiences. The issue of gender roles also underscores deep-rooted taboos, where patriarchal values dominate, often limiting women's participation in public life and reinforcing traditional roles within the home. Additionally, caste discrimination persists subtly within various communities, despite being formally abolished, leading to social divisions that restrict mobility and opportunities for individuals from marginalized groups. The collective hesitation to address these topics has created a culture of silence, where critical social issues are swept under the rug, hindering progress and reform. While a growing number of activists, particularly on social media, are challenging these taboos, the path to societal acceptance and open dialogue remains long and fraught with resistance.

26. Which of the following is highlighted as a taboo that hinders individuals from seeking help?

A. Mental health issues
B. Patriarchal values
C. Gender roles
D. Caste discrimination

27.What role do social media activists play according to the paragraph?

A. Reinforcing traditional taboos
B. Highlighting caste-based discrimination
C. Challenging societal taboos and promoting dialogue
D. Ignoring social issues to avoid backlash

28. What is the central theme of the paragraph?

A. Social activism in Pakistan
B. Caste-based discrimination in South Asia
C. Gender roles in Pakistani culture
D. Taboos affecting societal progress in Pakistan

Urdu literature holds a treasured place in South Asian cultural heritage, weaving together history, philosophy, romance, and social critique. Known for its rich poetry and prose, Urdu literature encompasses various forms, including ghazals, nazms, novels, and short stories, each contributing uniquely to its tapestry. The poetry of classical writers like Mirza Ghalib, Allama Iqbal, and Faiz Ahmed Faiz remains timeless, capturing universal human experiences while offering sharp insights into society and spirituality. Similarly, contemporary Urdu literature continues to address themes of identity, migration, and social justice, making it relevant for new generations. However, a debate persists regarding the accessibility of Urdu literature: while it holds immense depth, its classical vocabulary and intricate expressions can pose challenges for readers, especially younger generations unfamiliar with older linguistic forms. Advocates argue that preserving traditional language and structure is crucial to maintaining the authenticity of Urdu literature, while others suggest modernizing the language to make it more approachable. This ongoing conversation reflects a dynamic tension between tradition and accessibility, crucial to the evolution of Urdu literature.

29. Who among the following is considered a classical Urdu poet known for his philosophical themes?

A. Jaun Elia
B. Allama Iqbal
C. Qurratulain Hyder
D. Saadat Hasan Manto

30. Which of the following is a prominent form of Urdu poetry known for its emotional and romantic themes?

A. Essay
B. Novel
C. Haiku
D. Ghazal

ANALYTICAL REASONING

Question No 43 – 47

The organizers of a speech tournament will give an awards banquet after the final round of debates for the two teams in that round: for the Model School, composed of Instructor M and speakers' N O, and P, and for the Oxford School, composed of Instructor Q and speakers R, S, and U. At the winners' table exactly seven places will be set, all along one side of the table and consecutively numbered one through seven. The assignment of sitting to these seven positions will be determined by the team scores from the final round of debates and must meet the following conditions:

All the speakers will be seated.
If one team wins, the Instructor of the winning team will sit in the first position and a speaker from that team will sit in the fourth position.
In case of a tie, neither Instructor will sit at the winners table, and Judge Z will sit in the fourth position.
No two speakers from the same team can sit next to each other.
N and U will not sit next to each other.

31. Which of the following could be an acceptable assignment for chairs one

through four, respectively?

	One	Two	Three	Four
A.	**M**	**Q**	**P**	**U**
B.	**M**	**N**	**U**	**P**
C.	**Q**	**N**	**S**	**R**
D.	**Q**	**U**	**O**	**S**
E.	**S**	**U**	**P**	**R**

32. If O and U sit, respectively, in the second and seventh positions, which of the following must sit in the first position?

A. M
B. N
C. Q
D. P
E. R

33. If U and P are, respectively, in the first and seventh positions, which of the following must sit in the fifth position?

A. N
B. O
C. S
D. Q
E. R

34. If N, S, and P sit, respectively, in the third, fifth, and sixth positions which of the following must sit in the seventh position?

A. U
B. R
C. Q
D. M
E. O

35. If Q and N sit in the first and third positions, respectively, which of the following can sit in the fourth position?

A. O
B. P
C. Z
D. Z
E. S

Question No 48 – 50

A chemist has exactly four unlabeled glasses containing colorless liquids standing on a laboratory shelf. The chemist knows that the only six liquids that the glasses could possibly contain are pure M, pure N, pure O, or any combination of two of these, and that no two glasses are filled with the same liquid. The only feasible way of testing for the identity of the liquids is to use

strips of a special white reactive paper that turns Green, Brown, or Purple, depending on which of certain liquids it is dipped in. The full table of color changes and of the triggers for those changes is given below:

Green	***Brown***	***Purple***
Pure N	**Pure M**	**Pure M combine with N**
N combine with O	**Pure O**	**M combine with O**
	M, N, and O Combine together	

36. If the liquid in the first glass tested turns the reactive paper Green, and if the liquid in the second glass tested turns the paper Purple, then a combination of some of the liquid from each of the first two glasses tested will turn the reactive paper:

A. Green
B. Purple
C. either Green or Brown
D. either Green or Purple
E. either Brown or Purple

37. If none of the four liquids turns the reactive paper Purple, each of the following must be the contents of one of the glasses EXCEPT

A. pure M
B. pure N
C. pure O
D. N combined with O
E. M, N, and O combined together

38. If the liquid in the first glass tested turns the reactive paper Green and the liquid in the second glass tested turns it Brown, and if a combination of some of the liquid from each of the first two glasses tested turns it Green, then-which of the following must be true?

A. The first glass tested contains pure N.
B. The first glass tested contains N combined with O.
C. The second glass tested contains pure M.
D. The second glass tested contains pure O.
E. The second glass tested contains N combined with O.

Question No 51 – 53

Mehmood Trucking Company trips always begin in one of the five cities served by Mehmood—Noshara, Okara, Pashwar, Queta, and Rahim yar khan—and always end in one of these cities.
Mehmood drivers travel from city to city only on the routes listed in the mileage table below.

Noshara—Okara: 100 miles **Noshara—Pashwar: 100 miles**
Noshara—Rahim yar khan: 200 miles **Pashwar—Queta: 200 miles**
Queta—Rahim yar khan: 300 miles.
The routes listed above have no stretch or road in common.
The maximum distance an Mehmood driver can travel without making an

overnight stop is 300 miles.
Overnight slops must be made in cities served by Mehmood.

39. Which of the following is a trip an Mehmood driver could make without an overnight stop?

A. From Noshara to Rahim yar khan to Queta
B. From Rahim yar khan to Queta to Pashawar
C. From Queta to Rahim yar khan to Noshara
D. From Noshara to Pashwar to Queta

40. Which of the following is a list of the cities to which Mehmood driver can travel from Pashwar, returning to Pashwar on the same day he or she left?

A. Noshara only
B. Okara only
C. Queta only
D. Both Noshara and Okara

41. Which of the following accurately describes the possibilities for overnight stops during trips by Mehmood drivers that begin in Okara and end Quaker-town after including two other cities served by Mehmood?

A. The trips can be made without an overnight stop.
B. It is not possible to make an overnight stop at Noshara.
C. Rahim yar khan is the only possible place to make an overnight stop.
D. If exactly two overnight stops are made on a trip, they can be made only at Pashwar and Rahim yar khan.

<u>Question 54 – 56</u>

Five friends — Ali, Bilal, Fatima, Zara, and Hasan — are seated in a row from left to right (1 to 5). Each friend has a different favorite color: Red, Blue, Green, Yellow, and Pink. They are seated according to the following clues:
Ali is not seated at either end of the row.
Zara's favorite color is Yellow and she is seated immediately to the left of the person whose favorite color is Pink.
Fatima, whose favorite color is Green, is seated at one of the ends.
Bilal is seated to the right of Ali.
The person seated at the far right end has the favorite color Red.

42. Who is seated in the middle?

A. Ali
B. Bilal
C. Fatima
D. Zara

43. What is Hasan's favorite color?

A. Yellow

B. Red
C. Green
D. Blue

44. Which color is associated with Bilal?

A. Blue
B. Green
C. Pink
D. Yellow

Question 57 – 60

Consider a small office network where five devices — Router (R), Switch (S), Server (Sv), PC1, and PC2 — are connected in a specific configuration. Each device can only communicate with other devices based on their direct connections. Use the following rules to answer the questions:

The Router (R) provides internet access to the entire network but can only connect to the Switch (S).
The Switch (S) serves as a central hub, allowing PC1, PC2, and Server (Sv) to connect to it directly or through other devices.
PC1 and PC2 are set up for direct communication with each other.
The Server (Sv) can only be accessed by devices connected through the Switch (S).

45. If PC1 wants to communicate with the Server (Sv), which device must be functioning?

A. Router (R)
B. PC2
C. Switch (S)
D. None

46. Which device is directly accessible from the Router (R)?

A. PC1
B. Server (Sv)
C. Switch (S)
D. PC2

47. Which device provides internet access to PC2?

A. Switch (S)
B. Router (R)
C. Server (Sv)
D. PC1

48. If the Switch (S) fails, which devices lose internet access?

A. Only Server (Sv)
B. Only PC1 and PC2

C. **Server (Sv), PC1, and PC2**
D. **None**

Question 61 – 64

A variety show producer is interviewing five entertainers in five consecutive interviews. Each entertainer interviews alone, and only once. The five entertainers are: two singers (a tenor and a soprano), a dancer, a juggler, and a actor. The interviews must be scheduled according to the following conditions:
The two singers cannot interview one after the other.
The juggler must interview immediately before a singer.
The actor must interview immediately before or immediately after the dancer.

49. If the actor interviews first, which of the following must be true?

A. **The tenor interviews fifth.**
B. **The soprano interviews third.**
C. **The juggler interviews fourth.**
D. **The dancer interviews immediately before the tenor.**
E. **The soprano interviews sometime earlier than the dancer.**

50. If the actor, the soprano, and the juggler interview one after the other, in that order, which of the following must be true?

A. **The tenor is the fifth of the five to interview.**
B. **The juggler is the third of the five to interview.**
C. **The actor is the first of the five to interview.**
D. **The actor interviews immediately after the dancer.**

51. If the tenor interviews first and the soprano interviews fifth, which of the following must be true?

A. **The actor interviews sometime after the juggler.**
B. **The actor interviews immediately after the dancer.**
C. **The juggler interviews sometime after the dancer.**
D. **The dancer interviews immediately before the juggler.**

52. If the juggler interviews sometime earlier than the dancer, a singer CANNOT interview in which of the following positions?

A. **First**
B. **Second**
C. **Third**
D. **Fourth**
E. **Fifth**

MATH

60. For a particular machine that produce beads, 29 out of every 100 beads it has a defect. A bead produce by machine will be selected at random. What is the probability of selecting a bead that has a defect?

A. 1/2900
B. 1/29
C. 29/100
D. 29/10

61. If $f(x) = 2x^3 + 6$, then what is $f^{-1}(-10)$?

A. 1.26
B. -2
C. -7.08
D. -2.52
E. 2.83

62. If 4 +x 2 = 12, what is the value of 16x + 8?

A. 40
B. 60
C. 56
D. 48

63. A rocket contained 467,000 kilograms (kg) of propellant before launch. Exactly 21 seconds after launch, 362,105 kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

A. 4995
B. 17243
C. 39481
D. 104895

64. How many four digit numbers can be formed using the digits {1, 3, 5, 7, 9} (repetition is not allowed)?

A. 160
B. 180
C. 260
D. 360
E. 300

65. A bag contain nine yellow balls, two white balls and four red balls. In how many ways can two balls be drawn at random?

A. $^{2}C_{16}$

B. $^{15}C_2$
C. $^{16}C_2$
D. $^{2}C_{15}$

66. In the linear function f, f (0) = 8 and f (1) = 12. Which equation defines f?

A. f (x) = 12x + 8
B. f (x) = 4x - 1
C. f (x) = 4x + 12
D. f (x) = 4x + 8
E. f (x) = 4x

67. The circle has center O, circumference 144π, and diameters PR and QS. The length of arc PS is twice the length of arc PQ. What is the length of arc QR?

A. 24π
B. 48π
C. 72π
D. 96π

68. For the function q, the value of q (x) decreases by 45% for every increase in the value of x by 1. If q (0) = 14, which equation defines q?

A. q (x) = 0.55(14)^x
B. q (x) = 1.45(14)^x
C. q (x)= 14(0.55)^x
D. q (x)= 14(1.45)^x

69. The number of partition values in case of quintiles cannot exceed:

A. 4
B. 1
C. 3
D. 2

70. If for an individual series, assumed mean, A=25, Σ dx = -21 for dx = X – A and N = 7 then mean of series is:

A. 6.57
B. -3
C. 22
D. 20

71. Sum of absolute deviation about median is:

A. One
B. Zero
C. Minimum
D. Maximum

72. Two random variable X and Y are said to be independent if:

A. E(XY) = 1
B. E(XY) = 0
C. E(XY) = E(X) E(Y)
D. E(XY) = any constant

73. A committee has 5 men and 6 women. What are the number of ways of selecting a group of eight persons?

A. 165
B. 185
C. 205
D. 215
E. 225

74. Which expression is equivalent to $(m^4q^4z^{-1})$ (mq^5z^3), where m, q, and z are positive?

A. $m^4q^{20}z^{-3}$
B. $m^5q^9z^2$
C. $m^6q^8z^{-1}$
D. $m^{20}q^{12}z^{-2}$

75. For symmetrical distribution, M_d = ± Q.D covers:

A. 99% of data
B. 95% of data
C. 68% of data
D. 50% of data

76. The point (x,53) is a solution to the system of inequalities in the xy-plane. Which of the following could be the value of x?

$$y > 14$$
$$4x + y < 18$$

A. -9
B. -5
C. 9
D. 5

77. $$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A. 0.05
B. 12.1
C. 20.0
D. 242.0

78. Given that P(A) = 1/3, P(B) = 1/4, P(A|B) = 1/6, then P(B|A) =?

A. 1/4
B. 3/4
C. 1/8
D. None of these

79. In a class of 25 students, 80% are passing the class with a grade of C or better. If two students are randomly selected from the class, what is the probability that neither student is passing with a grade of C or better?

A. 0.03
B. 0.20
C. 0.08
D. 0.64
E. 0.63

80. If $2^x = 5$, then 5^x =?

A. 2
B. 2.32
C. 11.61
D. 41.97
E. 25

81. (x+y+3) (x+y-3)

A. $(x+y)^2 + 6(x+y) +9$
B. $(x+y)^2 + 6(x+y)$
C. $(x+y)^2 - 9$
D. $x^2 + y^2 – 3^2$

82. If 1 and −4 are both roots of a given polynomial, then which of the following must be a factor of the polynomial?

A. x + 1
B. $x^2 – 3x – 4$
C. $x^2 + 3x – 4$
D. x – 4
E. $x^2 – 1$

83. If f(x) = 4x − 1 and f (g (6)) = 1/3, then g(x) could equal which of the following?

A. 1/X
B. x^2
C. x/6
D. –x/18
E. 2/x

84. Which of the following could be a quadratic equation with integral coefficients having roots 3 + i and 3 − i?

A. $x^2 + 9x - 10 = 0$
B. $x^2 - 6x + 10 = 0$
C. $x^2 - 9x + 8 = 0$
D. $x^2 - 9 = 0$
E. $x^2 + 9x - 8 = 0$

85. In how many ways can the letters of the word NUMBER be arranged using all of the letters?

A. 720
B. 120
C. 24
D. 16

86. The second term of a geometric sequence is 6 and the 5th term is What is the common ratio of the sequence?

A. 3/2
B. 27/8
C. 2/3
D. $\sqrt[3]{6/4}$

87. If $x^2 - 5x + 1 = (x - a)\ 2 + c$, then what is the value of c?

A. 1
B. -29/4
C. -5/2
D. -21/4
E. -25/4

88. The first three terms of an arithmetic sequence are 3n, 6n − 2, and 8n + 1 where n is any real number. What is the value of the fourth term of the sequence?

A. 58.
B. 56
C. 54
D. 41

89. Which of the following is NOT a factor of $x^4 - 3x^3 - 11x^2 + 3x + 10$?

A. x + 1
B. x − 1
C. x + 2
D. x − 2
E. x − 5

90. An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

A. 9
B. 96
C. 120
D. 972

91. The mean age of 21 students in a biology class was 16.20 years. When a new student enrolled in the class, the mean age increased to 16.27 years. What was the age of the new student?

A. 16.3
B. 15.5
C. 16.5
D. 17.17

92. In how many ways can 12 people be divided into two groups if one group has 8 people and the other has 4?

A. 40320
B. 495
C. 1485
D. 11880

93. If each score in a set of scores is decreased by 2, which of the following would be true statements?

I. The mean is decreased by 2.
II. The mean is unchanged.
III. The standard deviation is decreased by 2.
IV. The standard deviation is unchanged.

A. I only
B. I and III only
C. II and IV only
D. I and IV only

94. The graph of f (x) = $|X|$ is translated 5 units down, 1 unit left, and reflected over the x-axis. If the resulting graph represents g(x), then g (−4) =

A. 2
B. −2
C. 4
D. 1

95. $c^{-1}(1/a + 1/b)$

A. 2/c
B. c((a+b)/ab)
C. (a+b)/abc
D. 1/abc

96. $\frac{10!}{3!7!}$

A. 1
B. 60
C. 120
D. 240

97. $\log_2 16\sqrt{2}$ **=?**

A. 9/2
B. 7/2
C. 4
D. 2

98. For all x such that x > 0, f (x) = $\log_3 x$**. What does** $f^{-1}(x)$ **equal?**

A. 3^x
B. x^3
C. $\log_3 x$
D. $\sqrt[3]{x}$

99. Seven integers are arranged from least to greatest. If the median is 9 and the only mode is 7, what is the least possible range for the 7 numbers?

A. 4
B. 5
C. 6
D. 8
E. 12

100. $2,500 is invested at a rate of 4.5% compounded monthly. The value of the investment in *t* years can be modeled by the equation $A = 2500\left(1 + \frac{0.045}{12}\right)^{12t}$**. How long will it take for the investment to double?**

A. 10.2 years
B. 15.4 years
C. 18.8 years
D. 25 years
E. 185.2 years